

Building Damage Classification and Streamflow Bias Correction via Deep Learning

CS 4440-101 — Artificial Intelligence — Final Project

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1 Introduction

This report presents two end-to-end machine learning pipelines: (1) **DoriaNET**—post-hurricane building damage classification from UAV aerial imagery—and (2) **Runoff Forecasting**—bias correction of NOAA National Water Model (NWM) short-range streamflow forecasts. Both tasks involve structured prediction under class imbalance or distributional mismatch, and each employs three models. Source code is available at <https://github.com/alanray/cs4440-final>.

2 Task 1: Building Damage Classification (DoriaNET)

2.1 Data and Pre-processing

The DoriaNET dataset [1] contains 1,458 building instances from three UAV videos over Marsh Harbour, Bahamas, captured 8 days after Hurricane Dorian (September 2019). Each instance is labeled with a FEMA HAZUS damage state (0–5), where 0 = no damage and 5 = destroyed/under construction. Class 3 (severe damage) dominates at 32%; class 5 (minority) accounts for 7% (Figure 1).

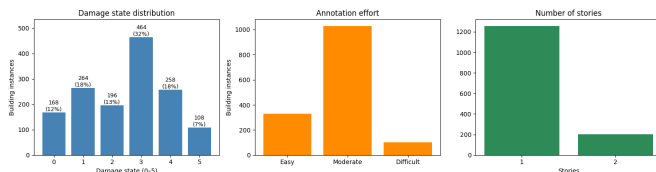


Figure 1: Dataset distributions: damage states (left), annotation effort (center), and number of stories (right). Class 3 dominates; annotation effort skews moderate.

A critical pre-processing choice distinguishes our pipeline from the original paper. Cheng et al. split on individual frame-instances, allowing the same physical building to appear in all three sets. We split on the 147 distinct physical buildings: all frame-appearances of a building stay together in train, val, or test. This **building-stratified split** eliminates data leakage and tests true generalization to unseen structures. As expected, our reported accuracy

is lower than the paper’s—consistent with their own cross-dataset results (30–40% exact accuracy), which represent the non-leaking comparison point.

Duplicate checking confirmed zero duplicate (building, frame) pairs. Splits were stratified by modal damage class and seed-searched (50 seeds) to maximize the smallest per-class cell: seed=8 yielded a minimum cell of $n = 23$. Final split: 794 train / 315 val / 349 test instances.

Per-building patches ($224 \times 224 \times 3$) were extracted by bounding-box-cropping the frame using the binary mask, padding to square, and resizing. Augmentation during training: random horizontal flip, ± 15 rotation, $\pm 20\%$ zoom. Pixel scaling: $[0, 1]$ for from-scratch models; MobileNetV2 preprocessor (scale to $[-1, 1]$) for transfer learning. Class imbalance was handled via cross-entropy alone; class-weighted loss destabilized training (model collapsed to predicting the highest-weighted class).

2.2 Models

Three architectures were implemented, tracking the course’s pedagogical progression from MLP to transfer learning (Table 1). Each consumes a $224 \times 224 \times 3$ patch and produces a 6-class softmax. Hyperparameter tuning was performed via random search (Keras Tuner, 20 trials) for the MLP and CNN; MobileNet variants used grid search over head size and augmentation.

Table 1: DoriaNET model architectures. MobileNetV1 and MobileNetV2 use ImageNet-pretrained weights with two-stage fine-tuning.

Model	Architecture	Member
MLP	Pool→Flatten→Dense(512)	Zach
CNN	3 conv blocks	Miguel
MobileNetV2	Pretrained + head	Alan

2.3 Evaluation and Results

Models were evaluated on the held-out test set using exact accuracy and ± 1 -class accuracy (a prediction within one damage level is considered correct—the standard for ordinal classification tasks like this). Training curves and confusion matrices are shown in Figures 2 and 3.

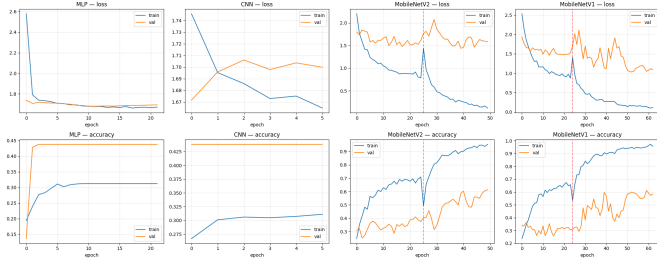


Figure 2: Training curves. MLP and CNN converge quickly but plateau near chance-level accuracy ($\sim 44\%$). MobileNet models continue improving and reach substantially higher accuracy via transfer learning.

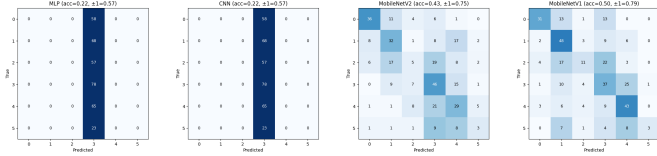


Figure 3: Confusion matrices on the test set. MLP and CNN collapse to predicting class 3 (the majority); MobileNetV2 and MobileNetV1 distribute predictions more broadly and attain meaningful per-class accuracy.

Table 2: DoriaNET test-set performance. MobileNetV1 achieves best exact and ± 1 accuracy; both transfer-learning models substantially outperform from-scratch baselines.

Model	Exact Acc.	± 1 Acc.
MLP	0.22	0.57
CNN	0.22	0.57
MobileNetV2	0.43	0.75
MobileNetV1	0.50	0.79

Both from-scratch models (MLP val accuracy 0.444, CNN 0.438) predict class 3 almost exclusively, confirming that without pretrained features and the small training set ($n = 794$), neither architecture can learn discriminative representations. Transfer learning reverses this: MobileNetV2 reaches 0.613 val accuracy, and MobileNetV1 reaches the same while generalizing slightly better on test (exact accuracy 0.50 vs. 0.43). The confidence calibration plot (Figure ??) shows MobileNetV2 has $R^2 = 0.816$ between confidence and accuracy, indicating well-calibrated uncertainty.

Confidence calibration ($R^2=0.816$ for MobileNetV2, 0.593 for MobileNetV1) shows MobileNetV2 is better calibrated despite lower absolute accuracy on the test set. Annotation-difficulty z-tests on MobileNetV1 found no statistically significant accuracy difference across easy/moderate/difficult strata ($p > 0.05$), likely due to small stratum sizes ($n_{\text{easy}}=58$, $n_{\text{difficult}}=43$).

3 Task 2: Streamflow Bias Correction (Runoff Forecasting)

3.1 Data and Pre-processing

NWM v2.1 short-range forecasts (lead times 1–18 h) and USGS hourly observations were collected for two stations: USGS 09520500 (Gila River near Dome, AZ; NHD reach 20380357) and USGS 11266500 (San Joaquin River, CA; reach 21609641). Data span April 2021–April 2023. Both series are in m^3/s (cms).

Pre-processing steps included: parsing NWM datetime strings with explicit format (`%Y-%m-%d.%H:%M:%S`) for speed; dropping the NWIS approval column; resampling USGS from 15-min to hourly via `asfreq('h')`; and forward/back-filling missing USGS values (0.4% at station 1; 5.8% at station 2). No duplicate instances were found.

Chronological split (required by spec): train Apr 2021–Jun 2022, val Jul–Sep 2022, test Oct 2022–Apr 2023. No test-set information influences training. The val window (summer baseflow) is unrepresentative of the test window, which includes the historic 2022–2023 California atmospheric river events; this drives the val $\rightarrow$ test MAE gap at station 2 (0.04 $\rightarrow$ 0.54 cms), not overfitting.

Features: [error, NWM_q , USGS_q] at $\text{lead}_h=1$, sampled hourly. Window length $L = 6$ hours (history), horizon $H = 18$ hours; input shape $(B, 6, 3)$. Inputs are z-scored (fit on train only); targets remain in raw cms. An L -sweep was performed (Figure 4); $L = 6$ matches Han & Morrison’s optimum.

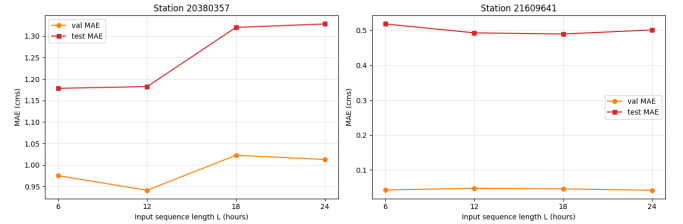


Figure 4: Validation and test MAE vs. input sequence length L for both stations. $L = 6$ minimizes val MAE at station 1; longer windows do not improve generalization.

Station 20380357 data context: the Gila River at this gauge is heavily regulated—Painted Rock Dam controls upstream releases, and essentially all natural flow is diverted for irrigation at the Ashurst-Hayden Diversion Dam. NWM models natural rainfall-runoff only; it does not simulate dams or diversions. The NWM–USGS correlation near zero ($\text{CC} \approx 0.04$) is a consequence of this physical decoupling, not a data error.

3.2 Models

Three recurrent architectures were trained per station (Table 3). Architecture depth was intentionally shallow,

following Han & Morrison’s published optimum. Loss: Huber (robust to flood-peak outliers). Optimizer: Adam, lr=1e-3, clipnorm=1.0. Early stopping on val_MAE (patience=20) with ReduceLROnPlateau.

Table 3: Runoff forecasting models.

Model	Params	Member
GRU	4,146	Alan Ray
LSTM simple	5,202	Zach Shotwell
LSTM stacked	13,522	Miguel Moreno Coin

Hyperparameter tuning: random search over GRU units $\in \{16, 32, 64\}$ and lr $\in \{10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ at station 21609641. Best: units=64, lr= 3×10^{-3} , val MAE 0.0419 (vs. 0.0447 for paper defaults). Training curves are shown in Figure 5.

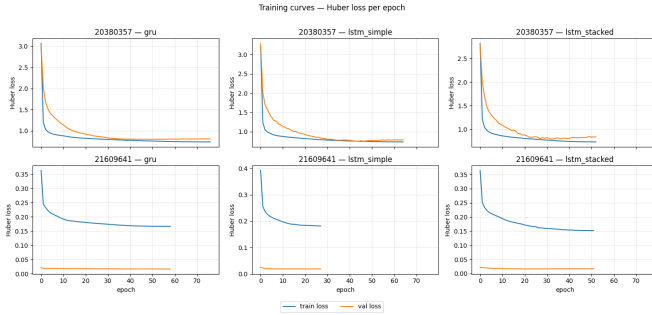


Figure 5: Huber loss training curves. Station 21609641 converges rapidly (val loss near zero). Station 20380357 shows larger train-val gap, consistent with the regulated/unregulated mismatch.

3.3 Evaluation and Results

Models are evaluated per lead time 1–18 h on CC, NSE, PBIAS, and RMSE, plus scatter R^2 . The persistence baseline (raw NWM) is the primary comparison.

Station 21609641 (California, large river): NWM was already well-calibrated (mean error +0.28 cms); the post-processor adds clear value at every lead time. GRU achieves $R^2=0.998$ at lead=1 h, NSE=0.997, and PBIAS reduced from -4.2% to -0.5% . The corrected forecast tracks 100+ cms atmospheric-river peaks accurately despite those events being absent from the val period (Figure ??). All three DL architectures perform within $\sim 5\%$ of each other (Table 4).

Station 20380357 (Arizona, regulated): NWM over-forecasted by a factor of $\sim 16\times$ (mean error -4.84 cms). Post-processors cut RMSE by $\sim 50\%$ ($8.4 \rightarrow 2.7$ cms at lead=1 h) and PBIAS by $\sim 10\times$, but Pearson CC remains near zero (≈ 0.08) and NSE stays deeply negative. This is consistent with the physical decoupling described above: the model learns a large bias correction but cannot recover dynamic skill where none exists in the input signal.

Table 4: Test-set performance at lead=1 h. GRU is best or tied-best at both stations. Sta. 2 metrics are excellent; Sta. 1 reflects the regulation/diversion mismatch.

Station	Model	CC	NSE	PBIAS	RMSE
20380357	Raw NWM	0.04	$\ll 0$	2130%	8.4
20380357	GRU	0.09	< 0	165%	2.7
20380357	LSTM simple	0.10	< 0	200%	2.9
20380357	LSTM stacked	0.08	< 0	375%	3.2
21609641	Raw NWM	0.995	0.989	-4.2%	1.73
21609641	GRU	0.999	0.997	-0.5%	0.86
21609641	LSTM simple	0.999	0.997	-0.8%	0.88
21609641	LSTM stacked	0.999	0.997	-1.0%	0.90

Time-series snippets confirm the corrected GRU forecast closely tracks observed flow during both a Dec 2022 atmospheric-river peak (~ 120 cms) and dry baseflow periods at station 21609641.

4 Conclusions

DoriaNET: Transfer learning is essential for this task. Without pretrained ImageNet features, both the MLP and CNN collapse to predicting the majority class under a building-stratified split. MobileNetV1 achieves 50% exact / $79\% \pm 1$ -class accuracy—competitive with the paper’s non-leaking cross-dataset baselines (30–40%). Future work: class-balanced sampling, ordinal loss functions, and integration with a building-detection front end.

Runoff forecasting: At a well-behaved, unregulated station, a shallow GRU with 4k parameters halves NWM’s RMSE and tracks extreme atmospheric river events ($R^2=0.998$ at lead=18 h). At a heavily regulated station, the post-processor successfully removes mean bias but cannot demonstrate dynamic skill—a consequence of NWM not modeling upstream dams and diversions. These two outcomes represent one unambiguous success and one identified data/model interaction that would require physics-informed inputs to resolve.

Across both tasks, building-stratified splitting and chronological splitting were non-negotiable methodological choices that produced more realistic evaluation metrics than the corresponding leaking alternatives.

References

- [1] Cheng, C. et al. (2021). *DoriaNET: A Dataset for Post-Hurricane Building Damage Assessment from UAV Imagery*. IEEE TGRS.
- [2] Han, M. & Morrison, R. (2022). *Streamflow Bias Correction via Data-Driven Post-Processing of NWM Forecasts*. Water Resources Research.